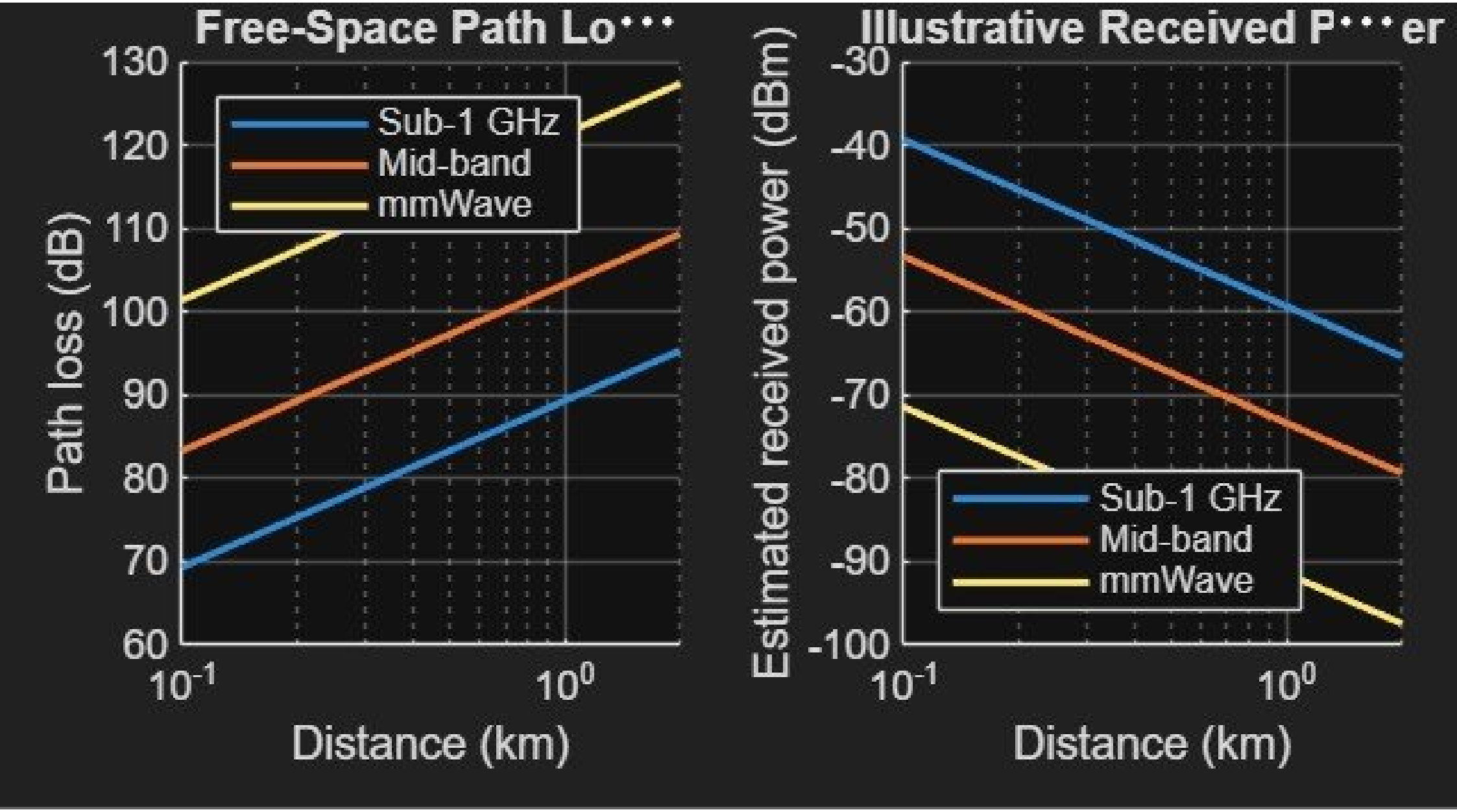
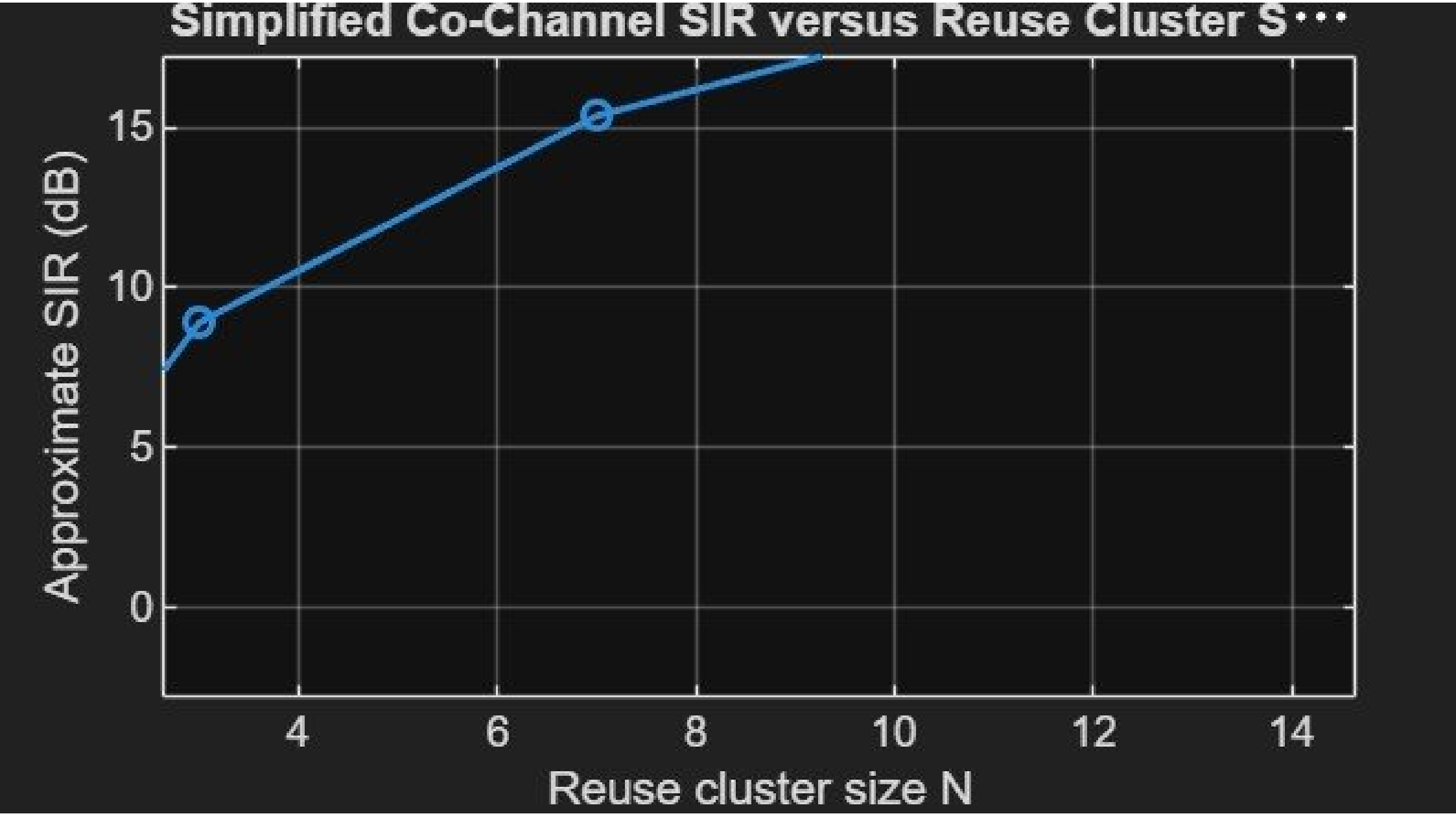
```
clear; clc; close all;
% Representative bands and link-budget assumptions
freqGHz = [0.7 3.5 28];
freqMHz = freqGHz * 1000;
bandName = ["Sub-1 GHz"; "Mid-band"; "mmWave"];
txPowerdBm = 30;
gtDbi = 0; grDbi = 0;
distanceKm = logspace(log10(0.1),log10(2),200);
distanceAtPointKm = 0.5;
% Free-space path loss (d in km, f in MHz)
fspl = @(fMHz,dKm) 32.44 + 20*log10(fMHz) + 20*log10(dKm);
lossAtPoint = fspl(freqMHz,distanceAtPointKm);
rxAtPoint = txPowerdBm + gtDbi + grDbi - lossAtPoint;
linkTable = table(bandName,freqGHz(:),lossAtPoint(:),rxAtPoint(:), ...
    'VariableNames',{'Band','Frequency_GHz','FSPL_dB','EstimatedRx_dBm'});
disp(linkTable);
% Path loss and received power over distance
figure; tiledlayout(1,2);
nexttile; hold on; grid on;
for k=1:numel(freqMHz)
    plot(distanceKm,fspl(freqMHz(k),distanceKm),'LineWidth',1.5);
end
set(gca,'XScale','log'); xlabel('Distance (km)'); ylabel('Path loss (dB)');
legend(bandName,'Location','northwest'); title('Free-Space Path Loss');
nexttile; hold on; grid on;
for k=1:numel(freqMHz)
    rx = txPowerdBm + gtDbi + grDbi - fspl(freqMHz(k),distanceKm);
    plot(distanceKm,rx,'LineWidth',1.5);
end
set(gca,'XScale','log'); xlabel('Distance (km)'); ylabel('Estimated received power (dBm)');
```

```
figure; tiledlayout(1,2);
nexttile; hold on; grid on;
for k=1:numel(freqMHz)
    plot(distanceKm,fspl(freqMHz(k),distanceKm),'LineWidth',1.5);
end
set(gca,'XScale','log'); xlabel('Distance (km)'); ylabel('Path loss (dB)');
legend(bandName,'Location','northwest'); title('Free-Space Path Loss');
nexttile; hold on; grid on;
for k=1:numel(freqMHz)
    rx = txPowerdBm + gtDbi + grDbi - fspl(freqMHz(k),distanceKm);
    plot(distanceKm,rx,'LineWidth',1.5);
end
set(gca,'XScale','log'); xlabel('Distance (km)'); ylabel('Estimated received power (dBm)');
legend(bandName,'Location','southwest'); title('Illustrative Received Power');
% Simple first-tier co-channel reuse estimate
cellRadiusKm = 0.5;
reuseN = [1 3 7 12];
pathLossExponent = 3.5;
reuseDistance = sqrt(3*reuseN) * cellRadiusKm;
sirLinear = (reuseDistance/cellRadiusKm).^pathLossExponent / 6;
sirDb = 10*log10(sirLinear);
reuseTable = table(reuseN(:),reuseDistance(:),sirDb(:), ...
    'VariableNames',{'ReuseClusterN','ApproxInterfererDistance_km','ApproxSIR_dB'});
disp(reuseTable);
figure; plot(reuseN,sirDb,'o-','LineWidth',1.5); grid on;
xlabel('Reuse cluster size N'); ylabel('Approximate SIR (dB)');
title('Simplified Co-Channel SIR versus Reuse Cluster Size');
fprintf('At %.2f km, estimated received powers are:\n',distanceAtPointKm);
disp(table(bandName,rxAtPoint(:),'VariableNames',{'Band','Rx_dBm'}));
```


Free-Space Path Lo•••
Path loss (dB)
130
120
110
100
90
80
70
60
Sub-1 GHz
Mid-band
mmWave
10^{-1}
10^{0}
Distance (km)
Illustrative Received P•••er
Estimated received power (dBm)
-30
-40
-50
-60
-70
-80
-90
-100
Sub-1 GHz
Mid-band
mmWave
10^{-1}
10^{0}
Distance (km)


Simplified Co-Channel SIR versus Reuse Cluster S
Approximate SIR (dB)
15
10
5
0
4
6
8
10
12
14
Reuse cluster size N

Command Window

```
    "Sub-1 GHz"          0.7          83.321          -53.321
    "Mid-band"           3.5          97.301          -67.301
    "mmWave"              28          115.36          -85.363

    ReuseClusterN    ApproxInterfererDistance_km    ApproxSIR_dB
    _____________    ___________________________    ____________

          1                        0.86603             0.56811
          3                            1.5              8.9177
          7                         2.2913              15.357
         12                              3              19.454

At 0.50 km, estimated received powers are:
        Band          Rx_dBm
    ____________    _______

    "Sub-1 GHz"     -53.321
    "Mid-band"      -67.301
    "mmWave"        -85.363

>>
```